Mohammed Anas W Z

Mini – Project Report
Computational Exploration of Single-Cell RNA-seq Data Using Seurat on Human Immune Cells

Introduction

In this project, I performed a comprehensive single-cell RNA sequencing (scRNA-seq) analysis using the Seurat workflow in R to characterize human immune cell populations. The analysis included data preprocessing, quality control (QC), dimensionality reduction, clustering, cell type annotation, differential expression analysis, and biological interpretation. The objective was to generate a robust immune cell atlas and interpret biologically meaningful patterns within the dataset.

The input dataset consisted of 32,357 genes across 10,937 cells in a single RNA assay.

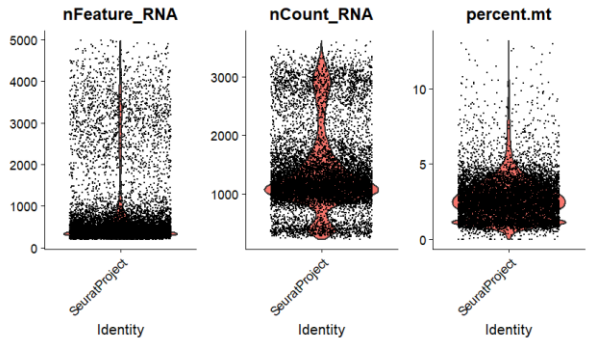
Quality Control and Filtering

I began by evaluating cell quality metrics including:

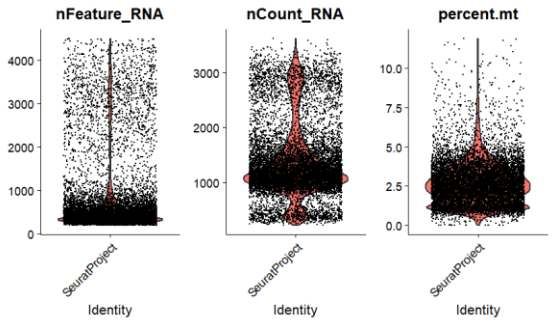
- Number of detected genes per cell (nFeature_RNA)
- Total UMI counts (nCount_RNA)
- Percentage of mitochondrial gene expression (percent.mt)

Based on QC violin and scatter plots (Figure 1 and Figure 2), I applied the following thresholds:

- nFeature_RNA > 200
- nFeature_RNA < 4500
- percent.mt < 12



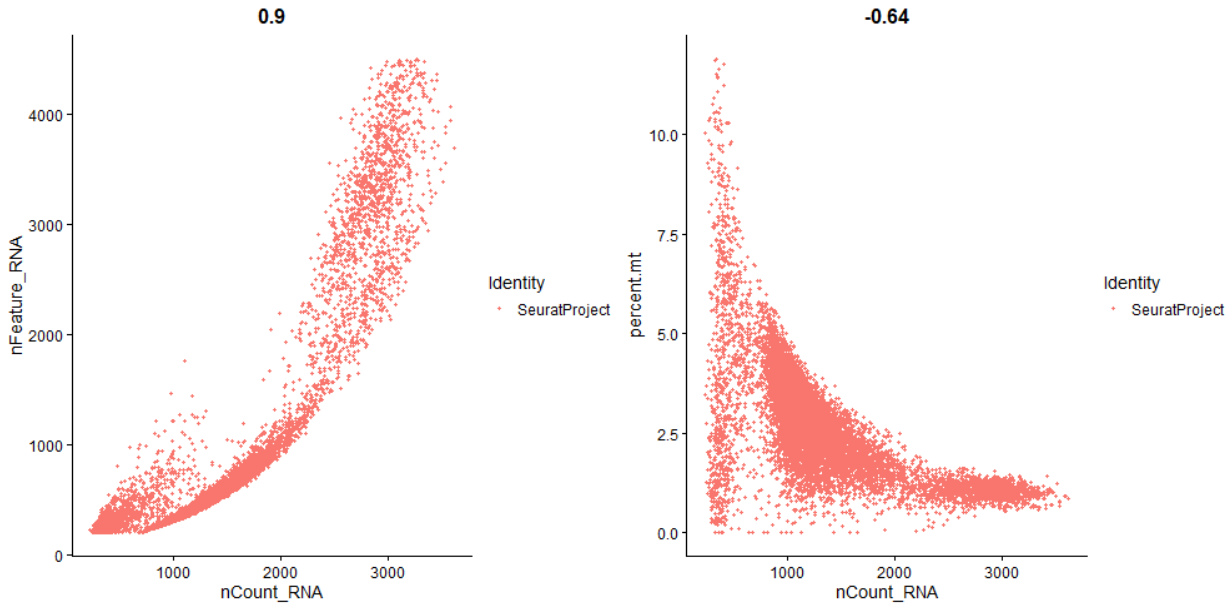
Before



After

Cells with fewer than 200 genes likely represented empty droplets or low-quality cells and were removed. Cells with more than 4500 genes showed unusually high complexity, suggesting potential doublets or multiplets. Additionally, cells with mitochondrial content above 12% were considered stressed or apoptotic and were excluded.

After filtering, 10,858 high-quality cells remained. Only 79 cells were removed, indicating that the dataset was generally of high quality and that filtering removed only clear outliers while preserving biological diversity.



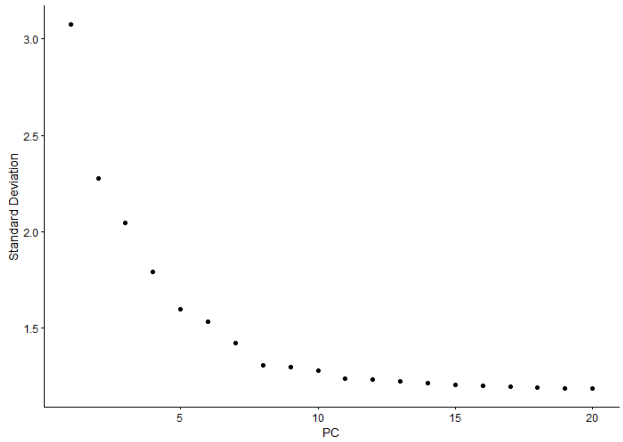
Dimensionality Reduction and Clustering

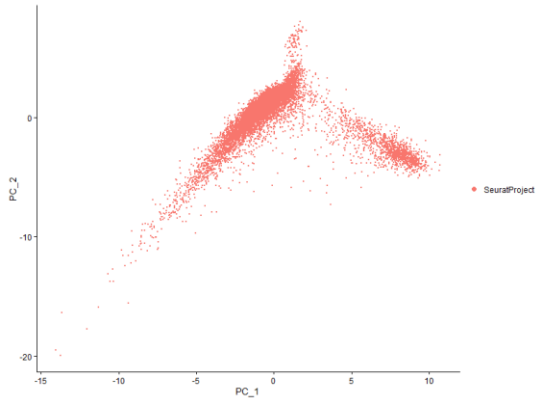
Next, I performed normalization, variable feature identification, scaling, and Principal Component Analysis (PCA). PCA diagnostics indicated that the first 15 principal components captured the majority of biologically relevant variance (Figure 3).

To assess cluster stability, I tested multiple PC ranges (1:10, 1:15, 1:20) and resolutions (0.3, 0.5, 0.8). The results were as follows:

PCs Used	Resolution 0.3	Resolution 0.5	Resolution 0.8
1:10	6 clusters	10 clusters	15 clusters
1:15	6 clusters	9 clusters	12 clusters

PCs Used	Resolution 0.3	Resolution 0.5	Resolution 0.8
1:20	6 clusters	10 clusters	12 clusters





At resolution 0.3, clustering consistently produced six major populations, indicating stable global structure. Increasing resolution to 0.8 resulted in over-clustering, splitting large clusters into small subclusters. Resolution 0.5 provided a balance between stability and biological interpretability.

Therefore, I selected:

- Dimensions: 1:15
- Resolution: 0.5
- Final clusters: 9

The UMAP visualization (Figure 4) showed clear separation between major immune populations.

Cell Type Annotation

To assign biological identities, I examined canonical marker gene expression using FeaturePlots and marker analysis (Figure 5). Based on gene expression patterns, I annotated clusters as follows:

Cluster 0 – Platelets
High expression of TREML1, CD9, MYL9, FLNA confirmed platelet identity.

Cluster 1 – Classical Monocytes ($CD14^+$)
Gene expression patterns consistent with homeostatic circulating monocytes.

Cluster 2 – Activated / Inflammatory Classical Monocytes
Strong expression of S100A8, S100A9, LYZ indicated inflammatory activation.

Cluster 3 – Erythrocytes
High expression of hemoglobin-related genes HBM, AHSP, SLC4A1, GYPB, CA1 clearly defined red blood cells.

Cluster 4 – Doublets / Contaminating Cells
Mixed or non-hematopoietic markers suggested technical doublets or contamination.

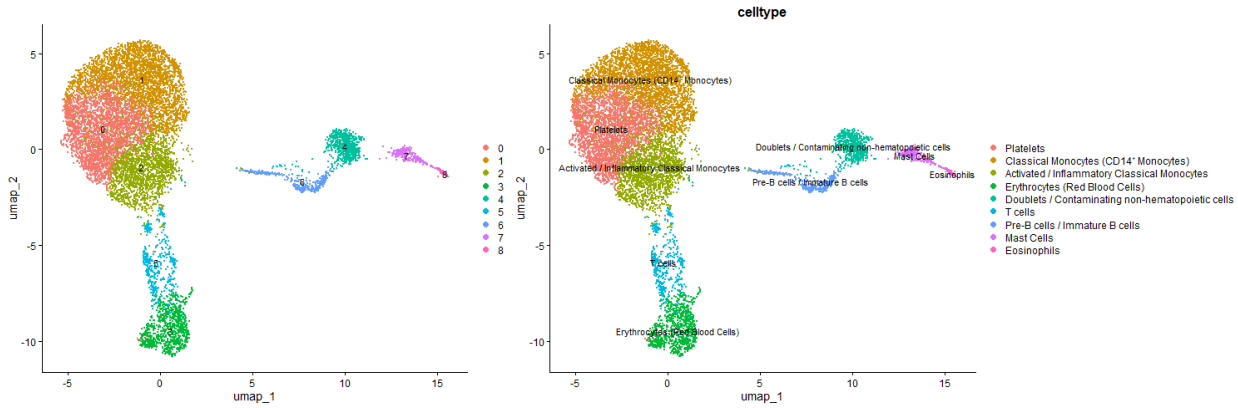
Cluster 5 – T Cells
Expression of CD2, CLEC2D, RORA confirmed lymphoid T cell identity.

Cluster 6 – Pre-B / Immature B Cells
Markers DNTT, VPREB1, MME (CD10) indicated early B cell lineage.

Cluster 7 – Mast Cells
High expression of TPSAB1, TPSB2, CPA3 confirmed mast cell identity.

Cluster 8 – Eosinophils
Strong expression of EPX, PRG2, CLC identified eosinophils.

These annotations were supported by canonical lineage-defining markers and consistent UMAP positioning.



Differential Expression Analysis

To explore functional heterogeneity within monocytes, I performed differential expression analysis between:

- Classical Monocytes (C1)

- Activated/Inflammatory Monocytes (C2)

Using:

- logfc.threshold = 0.25
- min.pct = 0.25

Activated monocytes (C2) showed strong upregulation of inflammatory genes including:

- S100A8
- S100A9
- LYZ

These genes are well-established mediators of innate immune activation and inflammation.

In contrast, classical monocytes (C1) exhibited genes associated with homeostatic immune surveillance rather than acute inflammatory response.

This analysis demonstrated a clear transcriptional shift from a resting to an activated inflammatory phenotype, confirming biologically meaningful clustering within the monocyte lineage (Figure 6).

Study-Linked Insight: Rare Granulocyte Populations

A key insight from this analysis was the identification of rare granulocyte populations:

- Mast Cells (C7)
- Eosinophils (C8)

Although these clusters contained relatively fewer cells compared to monocytes or T cells, they formed distinct UMAP clusters with highly specific markers.

Mast cells expressed TPSAB1, TPSB2, CPA3, which encode granule-associated proteases.

Eosinophils expressed EPX, PRG2, CLC, characteristic of type 2 immune responses.

The ability to resolve these low-frequency populations highlights the sensitivity of scRNA-seq in capturing rare immune subsets that may be masked in bulk RNA sequencing.

Overall Interpretation

Through systematic quality control, stable clustering, marker-based annotation, and differential expression analysis, I successfully constructed a detailed immune cell landscape from single-cell transcriptomic data.

The dataset revealed:

- Major immune populations (monocytes, T cells, B cells)
- Functional heterogeneity within monocytes
- Rare granulocyte subsets
- Technical doublets
- Non-immune components (erythrocytes, platelets)

The selected parameters (Dims 1:15, Resolution 0.5) provided a stable and biologically interpretable clustering solution.

Conclusion

In this project, I performed a complete Seurat-based scRNA-seq workflow from raw data to biological interpretation. By applying rigorous QC filtering, optimizing clustering parameters, annotating cell types using canonical markers, and performing differential expression analysis, I generated a comprehensive immune cell atlas.

The analysis demonstrated the power of single-cell RNA sequencing to resolve cellular heterogeneity, detect rare populations, and reveal functional states within immune lineages. This workflow provides a reproducible framework for future single-cell immune profiling studies.